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Threat Model

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Method	STRIDE, applied per data flow diagram element

1. Scope

This document analyses the initial release of NOVA: a locally deployed, single-user system with a multi-tenant-capable data model, no authentication, and no outbound network dependency beyond a host-resident inference runtime.

Controls are separated into those implemented for this release and those a production deployment would additionally require. Asserting production-grade security for a local single-user application would itself be a defect in this document.

Not claimed. This analysis does not assert conformance to any control framework. It does not claim that prompt injection is prevented, only that its blast radius is architecturally bounded.

2. System description

NOVA accepts natural-language requirements, generates structured test plans under schema constraint, and stores rules derived from user corrections. Stored rules are retrieved and injected into subsequent generation prompts.

The property that dominates this analysis: **a stored memory is replayed into every future prompt that retrieves it.** A compromise of the memory path is persistent, not transient.

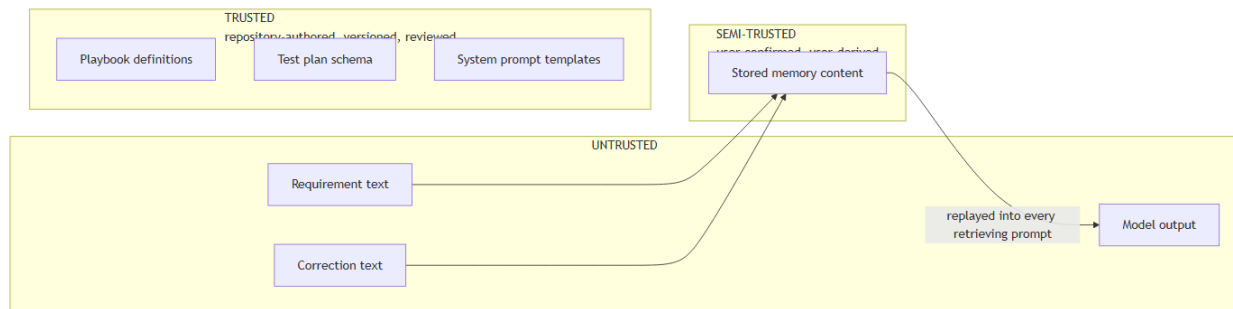
Architecture: [NOVA-SAD-001](#).

3. Assets

ID	Asset	Sensitivity	Rationale
A-1	Stored memories	High	Determine all future output. Compromise persists across every subsequent task
A-2	Prompt and configuration versions	High	Control system behaviour directly
A-3	Hosted provider credentials	High	Financial exposure and a data egress path
A-4	Requirement text	Medium, High with authentic data	May describe unreleased functionality
A-	Execution traces	Medium	Contain prompt context and retrieved memory

ID	Asset	Sensitivity	Rationale
5			content
A-6	Feedback events	Medium	The authoritative learning record. Corruption propagates to every derived score
A-7	Evaluation dataset	Low	Public by design

4. Trust boundaries



Boundary	Crossing	Control
TB-1	Browser to API service	Schema validation on every field. Tenant resolved server-side and never accepted from the client
TB-2	Untrusted text into a generation prompt	Delimited and labelled as data. Never concatenated into instruction context
TB-3	Model output into persistence or presentation	Schema conformance, then semantic validation, then escaping on render
TB-4	Correction text into durable memory	The highest-risk crossing. Mechanical screen, then mandatory human confirmation
TB-5	Tenant to tenant	Scoping applied at the data access layer, not at call sites
TB-6	Host to network	Local inference by default. A hosted provider is reachable only through an explicit user action

5. Threat enumeration

Threats are enumerated per STRIDE category and mapped to the boundary they cross.

5.1 Spoofing

ID	Threat	Boundary	Assessment
T1	A client asserts a tenant identity it does not hold	TB-1, TB-5	Mitigated. The tenant identifier is resolved server-side and never read from a request body, header, or parameter

ID	Threat	Boundary	Assessment
T2	An actor impersonates the legitimate user	TB-1	Accepted. No authentication exists in this release. The service binds to the loopback interface and the threat model assumes host access equates to user access. Authentication is required before any multi-user deployment

5.2 Tampering

ID	Threat	Boundary	Assessment
T3	Persistent injection through stored memory. Correction text shaped as a system instruction survives extraction, is confirmed, and is thereafter replayed into every retrieving prompt	TB-4	Mitigated by two independent controls in series, and by the absence of any capability to misuse. See section 6.1. Highest-impact threat in the system
T4	Injection through requirement text redirects generation	TB-2	Mitigated. Delimited and labelled untrusted blocks, constrained decoding bounding the output shape, output validation
T5	Adversarial feedback alters strategy selection	TB-1	Mitigated. Bounded update step, clamped score range, minimum observation threshold, per-tenant score isolation. No single event can alter a selection
T6	Model output containing markup, path traversal, or query-shaped text is interpreted rather than treated as data	TB-3	Mitigated. Schema conformance, semantic validation, parameterized queries only, escaping on render
T7	Direct modification of the datastore	Host	Accepted. Host access is assumed equivalent to user access in a local deployment

5.3 Repudiation

ID	Threat	Boundary	Assessment
T8	The origin of a stored memory cannot be established	TB-4	Mitigated. Provenance recorded against every memory: originating task, correction, timestamp. Feedback events are immutable and retained independently of any memory derived from them
T9	A configuration change cannot be attributed or reconstructed	TB-1	Mitigated. Versioned configuration records, retained on supersession, each activation linked to its evaluation run

5.4 Information disclosure

ID	Threat	Boundary	Assessment
T10	Cross-tenant disclosure through an unscoped query or an insecure direct object reference	TB-5	Mitigated. Scoping applied at the data access layer such that no query path can omit it. Verified by an unconditional gate. Admits no tolerance
T11	Requirement or memory content is transmitted off-host through a provider change	TB-6	Mitigated. Provider change requires explicit user action and never occurs as a fallback. The active provider is displayed in the interface
T12	Content is disclosed through application logs	Host	Mitigated. Identifiers and prompt digests only. Content logging exists solely as an explicit debug mode, and its absence is asserted by test
T13	Deleted memory content is recoverable from historical traces	TB-4	Mitigated. Traces store prompt digests, not prompt text. Deletion purges content and retains only a tombstone
T14	Credentials are disclosed through the repository or logs	Host	Mitigated. Environment file excluded from version control, secret scanning enabled. The default configuration requires no credential

5.5 Denial of service

ID	Threat	Boundary	Assessment
T15	An oversized requirement exhausts context or memory	TB-1	Mitigated. Input length caps, context budget guard reducing retrieved memory before requirement content
T16	A repeatedly failing job consumes resources without bound	Internal	Mitigated. Bounded attempt count, then terminal failure. Every retry loop bounded with a defined terminal state
T17	Unbounded memory growth degrades retrieval	Internal	Mitigated. Near-duplicate detection, confidence decay, archival below a floor, per-tenant cap
T18	Unbounded trace growth consumes storage	Internal	Accepted for this release. No pruning policy exists. Recorded as technical debt rather than discovered later

5.6 Elevation of privilege

ID	Threat	Boundary	Assessment
T19	An injected instruction causes the model to take an action beyond	TB-2, TB-4	Mitigated architecturally. The model has no tool surface: no shell, no filesystem, no network capability. A successful injection can produce a poor test case. It

ID	Threat	Boundary	Assessment
	generating text		cannot take an action, because no action exists
T20	A configuration version is activated without evaluation	TB-1	Mitigated. Activation without a linked passing evaluation run is rejected at the data layer, not merely in the interface. An interface-only check is advisory
T21	A malicious dependency executes code	Supply chain	Partially mitigated. Pinned versions with a lockfile, automated advisories, dependency audit in continuous integration, base images pinned by digest. Deliberate minimization of dependency count

6. Mitigations

6.1 The memory path, in detail

Threat T3 receives disproportionate treatment because its impact is persistent.

Layer	Control	Known weakness
1	Length bound on candidate rule text	Bounds payload size only
2	Generality, actionability, and scope-soundness checks	Rejects malformed candidates, not adversarial ones
3	Instruction-shape screen	A denylist against an open-ended attack surface. Will miss cases
4	Near-duplicate and conflict detection	Surfaces contradiction, does not detect intent
5	Mandatory human confirmation	Depends on the user reading the candidate. Degrades under high volume
6	No tool surface	The strongest layer, and architectural rather than filter-based
7	Adversarial regression suite	Establishes a floor, not a guarantee

Layers 3 and 5 are deliberately paired. Layer 3 is known to be incomplete, which is precisely why layer 5 exists behind it. Treating either as sufficient would be an error.

Layer 6 is what bounds the consequence. Absent a tool surface, the worst outcome of a successful injection is a poor test case that a human reviews.

6.2 Permission model

Principal	Capability
Web application	Only what the API exposes. No direct datastore access
API service	Datastore read and write within its tables. Inference gateway
Worker	Same datastore scope. Inference gateway. No inbound listener

Principal	Capability
The inference model	None. Generates text into a schema. Cannot read files, execute commands, or initiate network requests
Evaluation harness	A separate datastore instance. Never touches working data

The absence of a tool surface is a designed property, not an unimplemented feature. Introducing one later would require a permission model, sandboxing, and an approval flow, which is why such capabilities are excluded from this release.

6.3 Input and output validation

Boundary	Control
API inbound	Typed models, length caps, enumerated constraints, unknown fields rejected
Requirement text	Length cap tied to the context budget, encoding normalization
Correction text	Length cap, instruction-shape screen
Model output	Schema conformance, semantic checks, referential integrity of criterion links
Memory content	Validation screen before proposal, human confirmation before storage
Rendered output	Framework escaping. No unsanitized markup
Datastore	Parameterized queries exclusively

6.4 Tenant isolation

The single guarantee admitting no tolerance.

- Tenant identifier present on every user-owned relation
- Scoping applied at the data access layer, so isolation does not depend on any caller remembering
- Tenant resolved server-side, never from a client-supplied value
- No endpoint returns an object without a scope check
- The isolation suite is an unconditional gate from the milestone in which memory first exists, before a second tenant is ever created, because the test is what preserves the guarantee as the code grows
- With authentication, row-level security is added as defence in depth

7. Residual risk

ID	Residual risk	Rationale for acceptance
RR-1	Prompt injection is bounded, not prevented	The threat class is unsolved industry-wide. The defensible claim is architectural: the agent possesses no capability to misuse. The adversarial suite establishes a floor, not a guarantee

ID	Residual risk	Rationale for acceptance
RR-2	No authentication exists	Single-user local deployment bound to loopback. Host access is assumed equivalent to user access. Required before any multi-user deployment
RR-3	Personally identifiable information is not detected	The release processes synthetic and open-source data only. This status changes should authentic data enter the system , at which point detection becomes a requirement rather than a gap
RR-4	No encryption at rest	Local disk, synthetic data. Volume and column-level encryption required for production
RR-5	Trace growth is unbounded	Acceptable at single-user scale. Recorded as technical debt
RR-6	Memory conflict resolution is manual	Does not scale beyond a moderate memory count. Acceptable at expected volume
RR-7	Persistent injection defences are unproven	Novel threat with no established benchmark. The adversarial suite is the only evidence and is described as such

8. Assumptions

ID	Assumption	Consequence if false
SA-1	Host access is equivalent to legitimate user access	The absence of authentication becomes exploitable
SA-2	The instruction-shape screen catches enough to keep human review tractable	Review fatigue degrades the effectiveness of layer 5
SA-3	Data remains synthetic and open-source	Personally identifiable information detection becomes mandatory
SA-4	No tool capability is introduced	This entire document requires rewriting rather than amendment
SA-5	Services remain bound to the loopback interface	Network exposure introduces threats not analysed here

9. Controls by release

Control	Initial release	Production would require
Authentication	Stubbed, single user	Federated identity or session authentication, multi-factor
Authorization	Data-layer scoping	Additionally row-level security in the datastore
Rate limiting	Documented, middleware seam present	Enforced per-tenant token bucket, tighter on inference-invoking endpoints

Control	Initial release	Production would require
Personally identifiable information	Not detected. Recorded gap	Detection with redaction before storage
Encryption at rest	None	Volume encryption plus column-level for sensitive fields
Secret management	Environment file	Managed secret store
Audit	Application-level append-only	Additionally tamper-evident storage
Backup	Manual, drilled once	Scheduled, offsite, point-in-time recovery
Monitoring	Local tracing	Alerting, service objectives, on-call rotation

10. Verification

Every threat maps to a test. The mapping is maintained in [NOVA-STP-001 section 4.1](#).

Threat group	Verification
T1, T10	Isolation suite as an unconditional gate. Insecure direct object reference attempts against every identifier-bearing endpoint
T3	Adversarial scenarios targeting extraction and confirmation specifically: instruction-shaped corrections, oversized rules, rules crafted to suppress required case types
T4	Adversarial scenarios against the generation prompt: direct instruction, instructions concealed within acceptance criteria, role-play framing, encoded payloads
T5	Poisoning scenarios asserting score clamping and that no single event alters a selection
T6	Output handling scenarios: markup, traversal strings, query-shaped text, oversized fields
T12	Automated assertion that no content field appears in log output
T13	Deletion completeness test asserting non-retrievability and absence from subsequent prompts
T15, T16	Fault injection: oversized requirement, poison job, provider unavailability
T20	Attempted activation of an unevaluated configuration version, expecting rejection
T21	Dependency audit in continuous integration

11. Environmental notes

Note	Detail
Service binding	Loopback only. No reason exists to expose a local single-user application to a network segment
Host inference	The inference runtime is host-resident rather than containerized. GPU passthrough on this platform adds operational friction without benefit. The container-to-host path

Note	Detail
	requires verification and documentation
GPU contention	Endpoint protection software on the development workstation holds GPU memory, reducing available VRAM below nominal. Recorded so that measurements taken here remain interpretable elsewhere

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft